Sample covariance matrices, Marcenko-Pastur law

Let $x_1, \dots, x_n$ be i.i.d. random vectors in $\mathbb{R}^d$, satisfying

$$\mathbb{E} x_i = 0, \quad \mathbb{E} x_i x_i^T = \Sigma \in \mathbb{R}^{d \times d}.$$

Here $\Sigma \in \mathbb{R}^{d \times d}$ is the <u>population covariance matrix</u>.

Classical estimator: $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n x_i x_i^T \in \mathbb{R}^{d \times d}$, <u>sample covariance matrix</u>

Q: Suppose $n, d \to \infty$. When does the ESD of $\hat{\Sigma}$ have a well-defined limit, and how does this depend on Σ and $\frac{d}{n}$?

Assumptions: (i) $\mathbb{E} x_i = 0$.

(ii) $\mathbb{E} x_i x_i^T = \Sigma$, where $\|\Sigma\|_{op} \le C$ a constant.

(iii) $\forall p \ge 0$, $\exists C_p > 0$ s.t. for any $t > 0$, deterministic $A \in \mathbb{R}^{d \times d}$,

$$\mathbb{P}\left[\left| x_i^T A x_i - \underbrace{\operatorname{Tr} A\Sigma}_{= \mathbb{E} x_i^T A x_i} \right| \ge t \right] \le C_p \left(\frac{\|A\|_F}{t} \right)^p$$

Example: $x_i = T z_i$, $\|T\|_{op} \le C$, $z_i = (z_{i1}, \dots, z_{id}) \in \mathbb{R}^d$ has independent entries w/ $\mathbb{E} z_{ij} = 0$, $\mathbb{E} z_{ij}^2 = 1$, $\mathbb{E}|z_{ij}|^k \le C_k$. Then

- $\mathbb{E} x_i = T \cdot \mathbb{E} z_i = 0$
- $\mathbb{E} x_i x_i^T = T \cdot \mathbb{E} z_i z_i^T \cdot T^T = T T^T = \Sigma$

- $\mathbb{P}\left[\left|x_i^T A x_i - \operatorname{Tr} A\Sigma\right| \geq t\right]$

$= \mathbb{P}\left[\left|z_i^T T^T A T z_i - \operatorname{Tr} T^T A T\right| \geq t\right]$

$\leq C_p \left(\frac{\|T^T A T\|_F}{t}\right)^p$ (Lecture 3)

$\leq C_p \|T\|_{op}^{2p} \cdot \left(\frac{\|A\|_F}{t}\right)^p \equiv C_p' \left(\frac{\|A\|_F}{t}\right)^p.$

Thm: Let $x_1, \dots, x_n \in \mathbb{R}^d$ be i.i.d. and satisfy assumptions (i)-(iii).

Suppose $n, d \to \infty$ such that

- $\frac{d}{n} \to \gamma \in (0, \infty)$.
- ESD of Σ converges weakly to a prob. distribution ν.

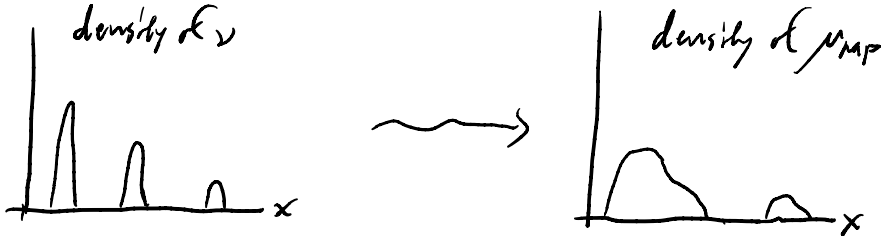
Then the ESD of $\hat{\Sigma}$ converges weakly a.s. to a prob. distribution μ_{MP} determined uniquely by (γ, ν):

For each $z \in \mathbb{C}^+$, there exists unique $\underline{m}(z) \in \mathbb{C}^+$ satisfying

$$z = -\frac{1}{\underline{m}(z)} + \gamma \int \frac{x}{1 + x\underline{m}(z)} \, d\nu(x) \qquad \text{(MP)}$$

Then μ_{MP} is the distribution w/ Stieltjes transform

$$m(z) = \frac{1}{\gamma} \underline{m}(z) + \left(1 - \frac{1}{\gamma}\right)\left(-\frac{1}{z}\right) \qquad (*)$$



Remarks: ① (*) is equivalently $\underline{m}(z) = \gamma m(z) + (1-\gamma)(-\frac{1}{z})$.

Substituting in (MP) and rearranging, (MP) is equivalent to

$$m(z) = \int \frac{1}{x(1-\gamma-\gamma z m(z)) - z} \, d\nu(x) \qquad \text{(MP')}$$

Thus $m(z)$ is the unique solution to (MP') in the domain $\{\frac{1}{\gamma}\underline{m} + (1-\frac{1}{\gamma})(-\frac{1}{z}) : \underline{m} \in \mathbb{C}^+\} \cap \mathbb{C}^+$.

(MP)/(MP') is often called the Marcenko-Pastur equation.

② If $\Sigma = I$, then ν is point mass at 1, so (MP) is

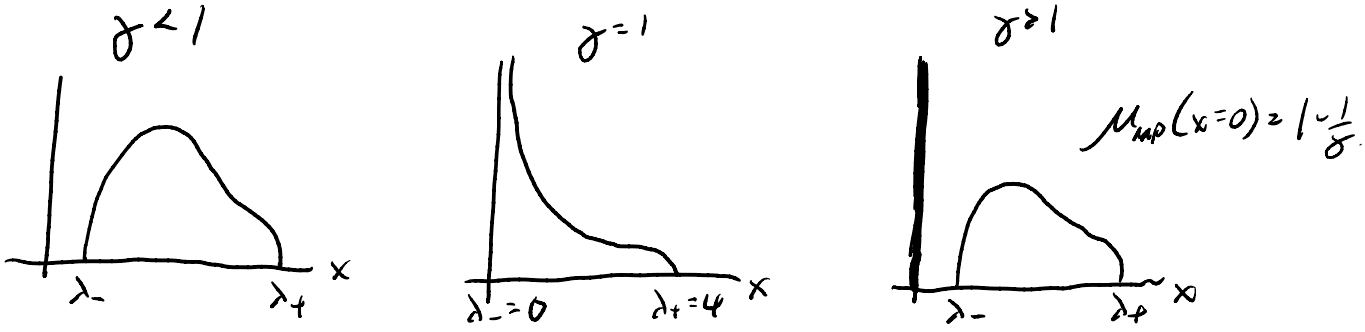
$$z = -\frac{1}{\underline{m}(z)} + \frac{\gamma}{1+\underline{m}(z)} \iff z\underline{m}(z)^2 + (1-\gamma+z)\underline{m}(z) + 1 = 0$$

Thus $\underline{m}(z) = \dfrac{-(1-\gamma+z) + \sqrt{(z-\lambda_+)(z-\lambda_-)}}{2z}$, $\lambda_\pm = (1 \pm \sqrt{\gamma})^2$

For each $x \in \mathbb{R} \setminus \{0\}$:

$$\lim_{z \in \mathbb{C}^+ \to x} \frac{1}{\pi} \operatorname{Im} m(z) = \begin{cases} \dfrac{1}{2\pi\gamma x}\sqrt{(\lambda_+ - x)(x - \lambda_-)} & \text{if } x \in (\lambda_-, \lambda_+) \\ 0 & \text{otherwise} \end{cases}$$

This defines the density of the Marcenko-Pastur law.



For general $\Sigma \neq I$ w/ limit ESD ν: μ_{MP} has a density

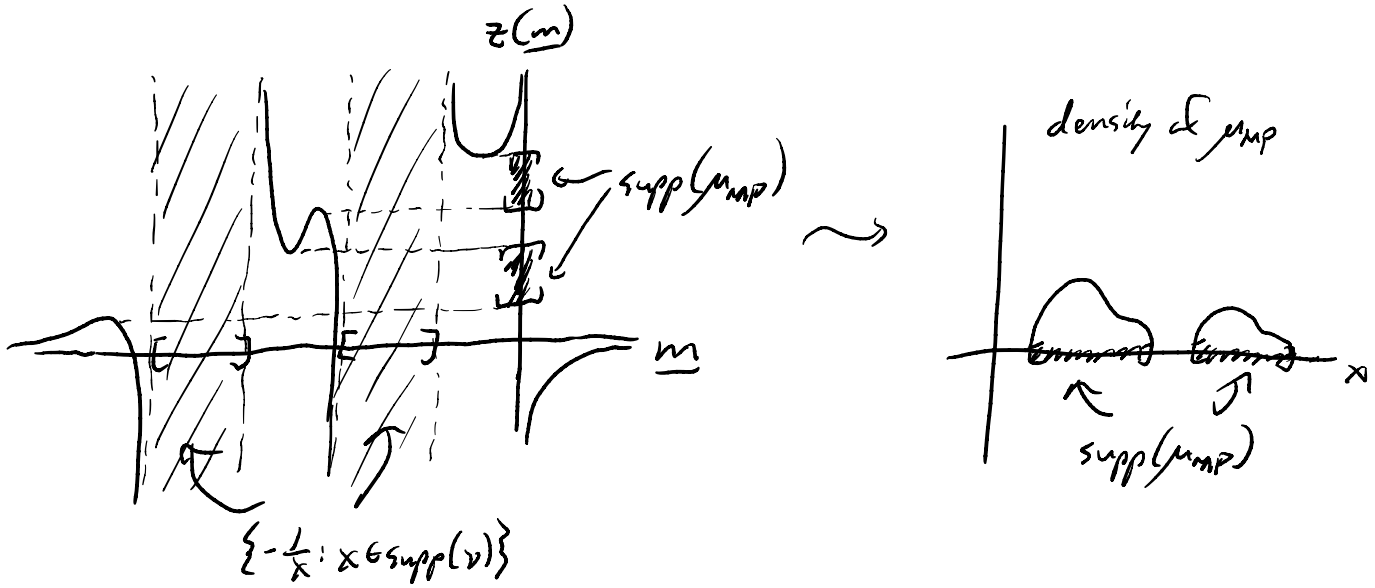
$$f(x) = \lim_{z \in \mathbb{C}^+ \to x} \frac{1}{\pi} \operatorname{Im} m(z) \quad \text{at each } x \in \mathbb{R} \setminus \{0\}.$$

There is no explicit formula for either $f(x)$ or $m(z)$.

However, there is a characterization of $\operatorname{supp}(\mu_{MP})$:

Let $z(\underline{m}) = -\frac{1}{\underline{m}} + \gamma \int \frac{x}{1 + x\underline{m}} d\nu(x)$, $\underline{m} \notin \{0\} \cup \{-\frac{1}{x} : x \in \operatorname{supp}(\nu)\}$

Then $\operatorname{supp}(\mu_{MP}) = \mathbb{R} \setminus \{ z(\underline{m}) : z'(\underline{m}) > 0 \}$.



[Silverstein, Choi '95]

(3) In the limit $\gamma \to 0$ (i.e. $n \gg d$), (MP') is

$$m(z) = \int \frac{1}{x - z} d\nu(x) + O(\gamma)$$

Thus $m(z) \to m_\nu(z) := \int \frac{1}{x - z} d\nu(x)$, so $\mu_{MP} \to \nu$ weakly.

I.e. the limit ESD of $\hat{\Sigma}$ approaches that of Σ.

(4) Let $X = [x_1 \cdots x_n] \in \mathbb{R}^{d\times n}$. Then $\hat{\Sigma} = \frac{1}{n} XX^T \in \mathbb{R}^{d\times d}$

Consider the Gram matrix $\underline{\hat{\Sigma}} = \frac{1}{n} X^T X \in \mathbb{R}^{n\times n}$.

If $X = UDV^T$ is SVD, $U \in \mathbb{R}^{d\times d}$, $V \in \mathbb{R}^{n\times n}$ orthogonal, $D \in \mathbb{R}^{d\times n}$,

then $\hat{\Sigma} = U\underbrace{\left(\frac{1}{n} DD^T\right)}_{\in \mathbb{R}^{d\times d}} U^T$, $\underline{\hat{\Sigma}} = V\underbrace{\left(\frac{1}{n} D^T D\right)}_{\in \mathbb{R}^{n\times n}} V^T$.

So eigenvalues of $\hat{\Sigma}$, $\underline{\hat{\Sigma}}$ are identical up to $|n-d|$ zeros.

Let $m_n(z)$, $\underline{m}_n(z)$ be Stieltjes transforms of their ESDs, let $\gamma_n = \frac{d}{n}$

$$\Rightarrow m_n(z) = \frac{1}{d}\sum_{i=1}^{d} \frac{1}{\lambda_i(\hat{\Sigma}) - z} = \frac{1}{d}\left(\sum_{i=1}^{n} \frac{1}{\lambda_i(\hat{\Sigma}) - z} + (d-n)\left(-\frac{1}{z}\right)\right)$$

$$= \frac{n}{d} \cdot \frac{1}{n}\sum_{i=1}^{n} \frac{1}{\lambda_i(\underline{\hat{\Sigma}}) - z} + \left(1 - \frac{n}{d}\right)\left(-\frac{1}{z}\right)$$

$$= \frac{1}{\gamma_n}\underline{m}_n(z) + \left(1 - \frac{1}{\gamma_n}\right)\left(-\frac{1}{z}\right)$$

Comparing w/ (*), the theorem shows $m_n(z) \to m(z)$ a.s. and

$\underline{m}_n(z) \to \underline{m}(z)$ a.s., i.e. $\underline{m}(z)$ is Stieltjes transform of ESD of $\underline{\hat{\Sigma}}$.

Proof of Marcenko-Pastur Theorem

Lemma: Let m_n, $\underline{m}_n$ be Stieltjes transforms of ESDs of $\hat{\Sigma}$, $\underline{\hat{\Sigma}}$.

For each fixed $z \in \mathbb{C}^+$, as $n, d \to \infty$,

$$z \cdot m_n(z) + \frac{1}{d} \operatorname{Tr}\left(I + \underline{m}_n(z)\Sigma\right)^{-1} \to 0 \quad \text{a.s.}$$

(Substituting $m_n(z) = \frac{1}{\gamma_n} \underline{m}_n(z) + \left(1 - \frac{1}{\gamma_n}\right)\left(-\frac{1}{z}\right)$ and taking $n, d \to \infty$ will yield the Marcenko-Pastur equation.)

Proof: Fix $z \in \mathbb{C}^+$. Let

$$R = (\hat{\Sigma} - zI)^{-1} = \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i^T - zI\right)^{-1}$$

$$R^{(j)} = \left(\frac{1}{n}\sum_{i \neq j} x_i x_i^T - zI\right)^{-1}$$

$$m_n = \frac{1}{d}\operatorname{Tr} R, \quad \underline{m}_n = \frac{d}{n} m_n + \left(1 - \frac{d}{n}\right)\left(-\frac{1}{z}\right) \left(= \frac{1}{n}\operatorname{Tr}(\hat{\Sigma} - zI)^{-1}\right)$$

Sherman-Morrison identity: $(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^TA^{-1}}{1 + v^TA^{-1}u}$

$$\Rightarrow R = R^{(j)} - \frac{\frac{1}{n}R^{(j)}x_j x_j^T R^{(j)}}{1 + \frac{1}{n}x_j^T R^{(j)} x_j}$$

$$\Rightarrow R x_j = R^{(j)} x_j - \frac{R^{(j)}x_j \cdot \frac{1}{n} x_j^T R^{(j)} x_j}{1 + \frac{1}{n} x_j^T R^{(j)} x_j} = \frac{R^{(j)} x_j}{1 + \frac{1}{n} x_j^T R^{(j)} x_j}$$

For any $M \in \mathbb{C}^{d \times d}$:

$$\frac{1}{d}\operatorname{Tr} M = \frac{1}{d}\operatorname{Tr} R(\hat{\Sigma} - zI)M \qquad \left(\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n} x_i x_i^T\right)$$

$$= -\frac{z}{d}\operatorname{Tr} RM + \frac{1}{d}\sum_{i=1}^{n} \frac{1}{n} x_i^T M R x_i$$

$$= -\frac{z}{d}\operatorname{Tr} RM + \frac{1}{d}\sum_{i=1}^{n} \frac{\frac{1}{n} x_i^T M R^{(i)} x_i}{1 + \frac{1}{n} x_i^T R^{(i)} x_i}$$

Choose $M = I$: $1 = -z \cdot m_n + \frac{1}{d}\sum_{i=1}^{n}\left(1 - \frac{1}{1 + \frac{1}{n} x_i^T R^{(i)} x_i}\right)$

$$= -z \cdot m_n + \frac{n}{d} - \frac{n}{d} \cdot \frac{1}{n}\sum_{i=1}^{n} \frac{1}{1 + \frac{1}{n} x_i^T R^{(i)} x_i}$$

$$\Rightarrow \frac{1}{n}\sum_{i=1}^{n} \frac{1}{1+\frac{1}{n}x_i^T R^{(i)} x_i} = 1 - \frac{d}{n} - \frac{d}{n}\cdot z\cdot m_n$$

$$= -z\left(\frac{d}{n}m_n + \left(1-\frac{d}{n}\right)\left(-\frac{1}{z}\right)\right) = -z\cdot \underline{m_n}$$

Heuristic calculation: $x_i^T M R^{(i)} x_i \approx \mathrm{Tr}\, M R^{(i)} \Sigma \approx \mathrm{Tr}\, R\Sigma M$

$$\text{So } \frac{1}{d}\mathrm{Tr}\, M \approx -\frac{z}{d}\mathrm{Tr}\, RM + \frac{1}{d}\mathrm{Tr}\, R\Sigma M \cdot \underbrace{\frac{1}{n}\sum_{i=1}^{n} \frac{1}{1+\frac{1}{n}x_i^T R^{(i)} x_i}}_{= -z\cdot \underline{m_n}}$$

$$\approx -\frac{z}{d}\mathrm{Tr}\, R(I + \underline{m_n}\Sigma)M$$

Lemma follows from choosing $M = (I + \underline{m_n}\Sigma)^{-1}$.

(1) Define $m_n^{(i)} = \frac{1}{d}\mathrm{Tr}\, R^{(i)}$, $\underline{m_n^{(i)}} = \frac{d}{n}m_n^{(i)} + \left(1-\frac{d}{n}\right)\left(-\frac{1}{z}\right)$.

Set $M = (I + \underline{m_n}\Sigma)^{-1}$, $M^{(i)} = (I + \underline{m_n^{(i)}}\Sigma)^{-1}$

$$\Rightarrow M - M^{(i)} = M(\underline{m_n^{(i)}} - \underline{m_n})\Sigma M^{(i)}$$

$$\Rightarrow \left|\frac{1}{n}x_i^T M R^{(i)} x_i - \frac{1}{n}x_i^T M^{(i)} R^{(i)} x_i\right|$$

$$\le \frac{1}{n}\left|\underline{m_n^{(i)}} - \underline{m_n}\right| \cdot \|M\|_{op}\cdot \|M^{(i)}\|_{op}\cdot \|R^{(i)}\|_{op}\cdot \|x_i\|_2^2.$$

For $z = \lambda + i\eta \in \mathbb{C}^+$, apply:

- $\|R^{(i)}\|_{op} \le \frac{1}{\eta}$
- For any $u \in \mathbb{R}^d$, symmetric p.s.d. $A = \sum_{k=1}^{d} \lambda_k v_k v_k^T \in \mathbb{R}^{d\times d}$:

$$\mathrm{Im}\, z\cdot u^T (A - zI)^{-1} u = \mathrm{Im}\, z \cdot \sum_{k=1}^{d} \frac{(u^T v_k)^2}{\lambda_k - z}$$

$$= \mathrm{Im} \sum_{k=1}^{d} \frac{(u^T v_k)^2 \cdot z(\lambda_k - \bar{z})}{|\lambda_k - z|^2} \geq 0$$

Thus: $|z(1 + \frac{1}{n} x_i^T R^{(i)} x_i)| \geq \mathrm{Im}\, z(1 + \frac{1}{n} x_i^T R^{(i)} x_i) \geq \mathrm{Im}\, z = \eta$

$$\Rightarrow |1 + \frac{1}{n} x_i^T R^{(i)} x_i| \geq \frac{\eta}{|z|}.$$

Also $\mathrm{Im}\, z m_n = \frac{1}{n} \sum_{i=1}^{n} \mathrm{Im}\, z \cdot e_i^T R e_i \geq 0$. Hence for any $t \geq 0$,

$$|z(1 + \underline{m}_n t)| \geq \mathrm{Im}\, z(1 + \underline{m}_n t)$$

$$= \mathrm{Im}\, z\left(1 + \left(\frac{d}{n} m_n + \left(1 - \frac{d}{n}\right)\left(-\frac{1}{z}\right)\right) t\right)$$

$$= \mathrm{Im}\, z + \frac{d}{n} \cdot t \cdot \mathrm{Im}\, z m_n \geq \mathrm{Im}\, z = \eta$$

$$\Rightarrow |1 + \underline{m}_n t| \geq \frac{\eta}{|z|}$$

$$\Rightarrow \|M\|_{op} = \max_{i=1}^{n} \frac{1}{|1 + \underline{m}_n \lambda_i(\Sigma)|} \leq \frac{|z|}{\eta}.$$

Similarly $\|M^{(i)}\|_{op} \leq \frac{|z|}{\eta}$.

- $|\underline{m}_n - \underline{m}_n^{(i)}| = \frac{d}{n} |m_n - m_n^{(i)}| = \frac{1}{n} |\mathrm{Tr}\, R - \mathrm{Tr}\, R^{(i)}| \leq \frac{2}{n\eta}$.

  by eigenvalue interlacing

- $\mathbb{P}\left[\left| \|x_i\|_2^2 - \mathrm{Tr}\, \Sigma \right| \geq t\right] \leq C_p \cdot \left(\frac{\|\Sigma\|_F}{t}\right)^p$

  where $\|\Sigma\|_F \leq \sqrt{d}$, $\mathrm{Tr}\, \Sigma \leq d \|\Sigma\|_{op} \leq Cn$

  $$\Rightarrow \max_{i=1}^{n} \|x_i\|_2^2 \leq C' n \quad \text{a.s. for all large } n.$$

Thus: $\left| \frac{1}{d}\sum_{i=1}^{n} \frac{\frac{1}{n}x_i^T M R^{(i)} x_i}{1+\frac{1}{n}x_i^T R^{(i)} x_i} - \frac{1}{d}\sum_{i=1}^{n} \frac{\frac{1}{n}x_i^T M^{(i)} R^{(i)} x_i}{1+\frac{1}{n}x_i^T R^{(i)} x_i} \right| \le \frac{C(z)}{n}$.

(2) Let $X^{(i)}$ denote $(x_1, \dots, x_n)$ removing x_i. Then $X^{(i)}$ is independent of x_i, and $M^{(i)}, R^{(i)}$ depend only on $X^{(i)}$, so

$$\mathbb{P}\left[\left| x_i^T M^{(i)} R^{(i)} x_i - \operatorname{Tr} M^{(i)} R^{(i)} \Sigma \right| \ge t \,\middle|\, X^{(i)} \right] \le C_p \left(\frac{\|M^{(i)} R^{(i)}\|_F}{t} \right)^p$$

where $\|M^{(i)} R^{(i)}\|_F \le \sqrt{d} \cdot \|M^{(i)}\|_{op} \cdot \|R^{(i)}\|_{op}$

$$\Rightarrow \sup_{i \ge 1} \left| x_i^T M^{(i)} R^{(i)} x_i - \operatorname{Tr} M^{(i)} R^{(i)} \Sigma \right| \le C(z) n^{1/2+\varepsilon} \text{ a.s. all large } n.$$

Thus: $\left| \frac{1}{d}\sum_{i=1}^{n} \frac{\frac{1}{n}x_i^T M^{(i)} R^{(i)} x_i}{1+\frac{1}{n}x_i^T R^{(i)} x_i} - \frac{1}{d}\sum_{i=1}^{n} \frac{\frac{1}{n}\operatorname{Tr} M^{(i)} R^{(i)} \Sigma}{1+\frac{1}{n}x_i^T R^{(i)} x_i} \right| \le C(z) n^{-1/2+\varepsilon}$ a.s. all large n.

(3) $\left| \frac{1}{n}\operatorname{Tr} M^{(i)} R^{(i)} \Sigma - \frac{1}{n}\operatorname{Tr} M R \Sigma \right|$

$$\le \left| \frac{1}{n}\operatorname{Tr}(M^{(i)} - M) R \Sigma \right| + \left| \frac{1}{n}\operatorname{Tr} M^{(i)} (R^{(i)} - R) \Sigma \right|$$

$$= \underbrace{\left| \frac{1}{n}\operatorname{Tr} M^{(i)} (m_n - m_n^{(i)}) \Sigma R \Sigma \right|}_{(A)} + \underbrace{\left| \frac{1}{n}\operatorname{Tr}(R^{(i)} - R) \Sigma M^{(i)} \right|}_{(B)}$$

- $(A) \le |m_n - m_n^{(i)}| \cdot \frac{d}{n} \|M^{(i)}\|_{op} \cdot \|R\|_{op} \cdot \|\Sigma\|_{op}^2 \le \frac{C(z)}{n}$.
- $\left| \operatorname{Tr}(R^{(i)} - R) \Sigma M^{(i)} \right| = \left| \operatorname{Tr} \frac{\frac{1}{n} R^{(i)} x_i x_i^T R^{(i)} \Sigma M^{(i)}}{1+\frac{1}{n}x_i^T R^{(i)} x_i} \right|$

$$\le \frac{\|\Sigma\|_{op} \cdot \|M^{(i)}\|_{op} \cdot \frac{1}{n}\|R^{(i)} x_i\|_2^2}{\left| 1+\frac{1}{n}x_i^T R^{(i)} x_i \right|}$$

For any $u \in \mathbb{R}^d$, symmetric $A = \sum_{k=1}^{d} \lambda_k v_k v_k^T \in \mathbb{R}^{d \times d}$:

$$\left| 1 + u^T (A - zI)^{-1} u \right| \geq \operatorname{Im}\left(1 + u^T (A - zI)^{-1} u\right) = \operatorname{Im} \sum_{k=1}^{d} \frac{(u^T v_k)^2}{\lambda_k - z}$$

$$= \operatorname{Im} \sum_{k=1}^{d} \frac{(u^T v_k)^2 (\lambda_k - \bar{z})}{|\lambda_k - z|^2} = \eta \cdot \sum_{k=1}^{d} \frac{(u^T v_k)^2}{|\lambda_k - z|^2} = \eta \, \| (A - zI)^{-1} u \|_2^2$$

$$\Rightarrow \left| \operatorname{Tr}\left(R^{(i)} - R\right) \Sigma M^{(i)} \right| \leq \frac{\|\Sigma\|_{op} \cdot \|M^{(i)}\|_{op}}{\eta}$$

$$\Rightarrow (B) \leq \frac{C(z)}{n}.$$

Thus: $$\left| \frac{1}{d} \sum_{i=1}^{n} \frac{\frac{1}{n} \operatorname{Tr} M^{(i)} R^{(i)} \Sigma}{1 + \frac{1}{n} x_i^T R^{(i)} x_i} - \underbrace{\frac{1}{d} \sum_{i=1}^{n} \frac{\frac{1}{n} \operatorname{Tr} M R \Sigma}{1 + \frac{1}{n} x_i^T R^{(i)} x_i}}_{= -z m_n \cdot \frac{1}{d} \operatorname{Tr} R \Sigma M} \right| \leq \frac{C(z)}{n}.$$

Combining (1) + (2) + (3):

$$\frac{1}{d} \operatorname{Tr} M z - \frac{z}{d} \operatorname{Tr} R M - z m_n \cdot \frac{1}{d} \operatorname{Tr} R \Sigma M + r_n$$

$$= -\frac{z}{d} \operatorname{Tr} R (I + m_n \Sigma) M + r_n = -\frac{z}{d} \operatorname{Tr} R + r_n,$$

where $|r_n| \to 0$ a.s. as $n, d \to \infty$. Hence

$$\frac{1}{d} \operatorname{Tr} R = m_n, \quad \frac{1}{d} \operatorname{Tr} M = \frac{1}{d} \operatorname{Tr} (I + m_n \Sigma)^{-1}$$

and rearranging shows the lemma. ∎

Proof of Thm: Fix $z = \lambda + i\eta \in \mathbb{C}^+$.

Let μ_n be ESD of $\hat{\Sigma}$. Let $\underline{\mu}_n$ be ESD of $\underline{\hat{\Sigma}}$.

Note: $\operatorname{Tr} \underline{\hat{\Sigma}} = \operatorname{Tr} \hat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} \|x_i\|_2^2 \le C_n$ a.s. for all large n,

by proof of previous lemma. Thus for any $D \ge 0$,

Thus $\underline{\mu_n}([D, \infty)) = \mathbb{P}_{X \sim \underline{\mu_n}}[X \ge D]$

$$\le \frac{1}{D} \mathbb{E}_{X \sim \underline{\mu_n}}[X] = \frac{\frac{1}{n} \operatorname{Tr} \underline{\hat{\Sigma}}}{D} \le \frac{C}{D}$$

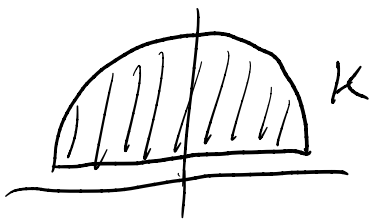
$\Rightarrow$ On event of probability 1, $\{\underline{\mu_n}\}_{n=1}^{\infty}$ and $\{\mu_n\}_{n=1}^{\infty}$ are tight.

Also $\operatorname{Im} \underline{m_n} = \operatorname{Im} \frac{1}{n} \sum_{i=1}^{n} \frac{1}{\lambda_i(\underline{\hat{\Sigma}}) - z} = \frac{1}{n} \sum_{i=1}^{n} \frac{\eta}{(\lambda_i(\underline{\hat{\Sigma}}) - \lambda)^2 + \eta^2}$

$$\ge \underbrace{\left(1 - \frac{C}{D}\right) \cdot \frac{\eta}{(D + \lambda)^2 + \eta^2}}_{:= \delta} \quad \text{a.s. all large } n.$$

$\Rightarrow$ On event of probability 1, for all large n

$$\underline{m_n} \in K := \left\{ m \in \mathbb{C}^+ : |m| \le \frac{1}{\eta}, \ \operatorname{Im} m \ge \delta \right\}$$



Let ν_n be ESD of Σ, $\gamma_n = \frac{d}{n}$. Lemma shows

$$\underbrace{z \cdot \underline{m_n}}_{= \frac{z}{\gamma_n} \cdot \underline{m_n} - \left(1 - \frac{1}{\gamma_n}\right)} + \underbrace{\int \frac{1}{1 + \underline{m_n} x} \, d\nu_n(x)}_{= 1 - \underline{m_n} \int \frac{x}{1 + \underline{m_n} x} \, d\nu_n(x)} \to 0 \quad \text{a.s.}$$

Multiply by $\frac{\gamma_n}{\underline{m_n}}$: $z + \frac{1}{\underline{m_n}} - \gamma_n \int \frac{x}{1 + \underline{m_n} x} \, d\nu_n(x) \to 0$ a.s.

Let $f_n(m) = z + \frac{1}{m} - \gamma_n \int \frac{x}{1+mx}\, d\nu_n(x)$

$f(m) = z + \frac{1}{m} - \gamma \int \frac{x}{1+mx}\, d\nu(x)$

Then $f_n(m) \to f(m)$ $\forall m \in K$, and f_n, f are uniformly Lipschitz continuous on K.

- Since K is compact, any subsequence has sub-subsequence $\{n_k\}_{k=1}^{\infty}$ where $\underline{m}_{n_k} \to \underline{m}$ for some limit $\underline{m} \in K$. Then $f_{n_k}(\underline{m}_{n_k}) \to 0$ and

$$|f(\underline{m}) - f_{n_k}(\underline{m}_{n_k})|$$
$$\leq |f(\underline{m}) - f_{n_k}(\underline{m})| + |f_{n_k}(\underline{m}) - f_{n_k}(\underline{m}_{n_k})| \longrightarrow 0$$

i.e. $f(\underline{m}) = 0$. Thus there exists $\underline{m} \in K \subseteq \mathbb{C}^+$ where

$$z = -\frac{1}{\underline{m}} + \gamma \int \frac{x}{1+x\underline{m}}\, d\nu(x) \qquad (MP)$$

If $f(\underline{m}) = f(\underline{m}') = 0$ for $\underline{m} \neq \underline{m}' \in \mathbb{C}^+$, then

$$0 = \frac{1}{\underline{m}} - \frac{1}{\underline{m}'} - \gamma \int \frac{x}{1+x\underline{m}}\, d\nu(x) + \gamma \int \frac{x}{1+x\underline{m}'}\, d\nu(x)$$

$$= \frac{\underline{m}' - \underline{m}}{\underline{m}\,\underline{m}'} - \gamma \int \frac{x^2(\underline{m}' - \underline{m})}{(1+x\underline{m})(1+x\underline{m}')}\, d\nu(x)$$

$$\Rightarrow 1 = \gamma \int \frac{x^2 \underline{m}\,\underline{m}'}{(1+x\underline{m})(1+x\underline{m}')}\, d\nu(x)$$

$$\leq \left(\gamma \int \left|\frac{x\underline{m}}{1+x\underline{m}}\right|^2 d\nu(x)\right)^{1/2} \left(\gamma \int \left|\frac{x\underline{m}'}{1+x\underline{m}'}\right|^2 d\nu(x)\right)^{1/2}$$

But $\Im(\underline{m}) > 0$ and $\operatorname{Im} \underline{m} > 0$ imply

$$0 < \operatorname{Im} z = \operatorname{Im}\left(-\frac{1}{\underline{m}} + \gamma \int \frac{x}{1+x\underline{m}}\, d\nu(x)\right)$$

$$= \frac{\operatorname{Im} \underline{m}}{|\underline{m}|^2} - \gamma \int \frac{x^2 \operatorname{Im} \underline{m}}{|1+x\underline{m}|^2}\, d\nu(x)$$

$$\Rightarrow \gamma \int \left|\frac{x\underline{m}}{1+x\underline{m}}\right|^2 d\nu(x) < 1, \quad \text{a contradiction.}$$

So there exists a unique value $\underline{m} \in \mathbb{C}^+$ satisfying (MP).

Then this is the limit for every subsequence, so $\underline{m}_n \to \underline{m}$ a.s. Then also

$$m_n(z) = \frac{1}{\gamma_n}\underline{m}_n + \left(1 - \frac{1}{\gamma_n}\right)\left(-\frac{1}{z}\right) \to m(z) = \frac{1}{\gamma}\underline{m} + \left(1-\frac{1}{\gamma}\right)\left(-\frac{1}{z}\right) \text{ a.s.}$$

- Since $\{\mu_n\}_{n=1}^{\infty}$ is tight, any subsequence has sub-subsequence $\{n_k\}_{k=1}^{\infty}$ where $\mu_{n_k} \to \mu$ weakly for some prob. measure μ.

Then $m_{n_k}(z) = \int \frac{1}{x-z}\, d\mu_{n_k}(x) \to \int \frac{1}{x-z}\, d\mu(x)$ $\forall z \in \mathbb{C}^+$,

by weak convergence. Then $m(z) = \int \frac{1}{x-z}\, d\mu(x)$, so this weak limit is the same for every subsequence and $m(z)$ is its Stieltjes transform. ∎

Problem 5: Suppose $x_1, \dots, x_n \overset{iid}{\sim} \mathcal{N}(0, \Sigma)$ where $\Sigma = \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_d \end{pmatrix}$ is diagonal, $\sigma_1, \dots, \sigma_d \in [c, C]$ for constants $C, c > 0$.

Let $g_i = \Sigma^{-1/2} x_i \sim \mathcal{N}(0, I)$, $G = [g_1 \cdots g_n] \in \mathbb{R}^{d \times n}$, $z \in \mathbb{C}^+$,

$$H(z) = \begin{pmatrix} -z I_{n \times n} & \frac{1}{\sqrt{n}} G^T \\ \frac{1}{\sqrt{n}} G & -\Sigma^{-1} \end{pmatrix}, \quad R(z) = H(z)^{-1} = \begin{pmatrix} R_{11} & R_{21}^T \\ R_{21} & R_{22} \end{pmatrix}$$

(a) Show $\underline{m}_n = \frac{1}{n} \operatorname{Tr} R_{11}$, $z \cdot m_n = \frac{1}{d} \operatorname{Tr} \Sigma^{-1} R_{22}$

(b) Applying Stein's lemma to the (α, α) coordinate of

$$I = \mathbb{E}[H(z) R(z)]$$

show that for each $\alpha = 1, \dots, d$,

$$1 = -\sigma_\alpha^{-1} \mathbb{E}[(R_{22})_{\alpha\alpha}] - \mathbb{E}[(R_{22})_{\alpha\alpha} \underline{m}_n] - \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[(R_{21})_{\alpha i}^2]$$

(c) Show that

$$\left| \mathbb{E}[(R_{22})_{\alpha\alpha} \underline{m}_n] - \mathbb{E}[(R_{22})_{\alpha\alpha}] \cdot \mathbb{E} \underline{m}_n \right| \le \frac{C(z)}{\sqrt{n}},$$

hence as $n, d \to \infty$,

$$\underbrace{\frac{1}{d} \sum_{\alpha=1}^{d} \sigma_\alpha^{-1} \mathbb{E}[(R_{22})_{\alpha\alpha}]}_{= \mathbb{E}[z \cdot m_n]} + \underbrace{\frac{1}{d} \sum_{\alpha=1}^{d} \frac{1}{1 + \sigma_\alpha \mathbb{E} \underline{m}_n}}_{= \frac{1}{d} \operatorname{Tr}(I + \mathbb{E} \underline{m}_n \cdot \Sigma)^{-1}} \to 0$$